Held-out DCI-C across all datasets

Gemma-2-2B

Dataset	SSAE L1	SSAE Step-L0	SSAE sigmoid-L0	SSAE ReLU	SSAE JumpReLU	SSAE TopK	SAE ReLU	SAE JumpReLU	SAE TopK
ENG→FR	0.672 ± 0.002	0.664 ± 0.003	0.661 ± 0.004	0.065 ± 0.005	0.933 ± 0.079	0.937 ± 0.023	0.355 ± 0.048	0.989 ± 0.015	0.982 ± 0.026
ENG→DE	0.736 ± 0.060	0.686 ± 0.018	0.640 ± 0.001	0.062 ± 0.003	0.911 ± 0.009	0.940 ± 0.044	0.238 ± 0.015	0.921 ± 0.057	0.870 ± 0.022
GENDER	0.836 ± 0.057	0.902 ± 0.070	0.847 ± 0.110	0.103 ± 0.006	0.811 ± 0.038	0.657 ± 0.047	0.255 ± 0.021	0.754 ± 0.033	0.741 ± 0.023
BINARY	0.628 ± 0.005	0.608 ± 0.002	0.679 ± 0.036	0.260 ± 0.002	0.635 ± 0.009	0.652 ± 0.027	0.210 ± 0.014	0.629 ± 0.029	0.583 ± 0.019
CORR	0.733 ± 0.034	0.753 ± 0.022	0.799 ± 0.027	0.123 ± 0.002	0.748 ± 0.032	0.777 ± 0.018	0.192 ± 0.009	0.709 ± 0.038	0.667 ± 0.013
TruthfulQA	0.536 ± 0.018	0.612 ± 0.047	0.608 ± 0.010	0.213 ± 0.005	0.570 ± 0.024	0.515 ± 0.035	0.387 ± 0.010	0.513 ± 0.029	0.434 ± 0.041
Sycophancy	0.608 ± 0.011	0.616 ± 0.001	0.573 ± 0.024	0.465 ± 0.021	0.578 ± 0.003	0.610 ± 0.025	0.356 ± 0.005	0.562 ± 0.011	0.444 ± 0.017
Refusal	0.581 ± 0.022	0.677 ± 0.035	0.718 ± 0.037	0.264 ± 0.015	0.665 ± 0.032	0.593 ± 0.033	0.409 ± 0.028	0.583 ± 0.024	0.429 ± 0.032
Bias-in-Bios	0.485 ± 0.006	0.391 ± 0.003	0.353 ± 0.005	0.575 ± 0.012	0.414 ± 0.001	0.623 ± 0.006	0.587 ± 0.004	0.421 ± 0.005	0.599 ± 0.008

Pythia-70M

Dataset	SSAE L1	SSAE Step-L0	SSAE sigmoid-L0	SSAE ReLU	SSAE JumpReLU	SSAE TopK	SAE ReLU	SAE JumpReLU	SAE TopK
ENG→FR	0.771 ± 0.102	0.936 ± 0.015	0.871 ± 0.054	0.052 ± 0.002	0.812 ± 0.053	0.432 ± 0.004	0.065 ± 0.003	0.619 ± 0.040	0.695 ± 0.046
ENG→DE	0.825 ± 0.044	0.826 ± 0.055	0.887 ± 0.002	0.050 ± 0.004	0.795 ± 0.063	0.446 ± 0.012	0.059 ± 0.007	0.627 ± 0.013	0.607 ± 0.041
GENDER	0.791 ± 0.112	0.942 ± 0.024	0.843 ± 0.078	0.060 ± 0.004	0.727 ± 0.055	0.527 ± 0.053	0.065 ± 0.005	0.623 ± 0.070	0.608 ± 0.011
BINARY	0.314 ± 0.004	0.354 ± 0.010	0.361 ± 0.034	0.250 ± 0.002	0.391 ± 0.021	0.275 ± 0.007	0.129 ± 0.010	0.323 ± 0.041	0.247 ± 0.023
CORR	0.355 ± 0.019	0.376 ± 0.027	0.408 ± 0.012	0.125 ± 0.003	0.363 ± 0.009	0.325 ± 0.008	0.174 ± 0.003	0.339 ± 0.007	0.320 ± 0.037
TruthfulQA	0.314 ± 0.007	0.304 ± 0.007	0.354 ± 0.020	0.299 ± 0.013	0.351 ± 0.011	0.349 ± 0.012	0.399 ± 0.012	0.327 ± 0.005	0.313 ± 0.007
Sycophancy	0.458 ± 0.040	0.357 ± 0.012	0.333 ± 0.013	0.345 ± 0.012	0.390 ± 0.007	0.290 ± 0.015	0.309 ± 0.015	0.294 ± 0.011	0.222 ± 0.021
Refusal	0.481 ± 0.039	0.492 ± 0.021	0.532 ± 0.026	0.397 ± 0.010	0.538 ± 0.054	0.505 ± 0.015	0.531 ± 0.011	0.530 ± 0.026	0.456 ± 0.017
Bias-in-Bios	0.248 ± 0.001	0.273 ± 0.004	0.214 ± 0.002	0.307 ± 0.005	0.251 ± 0.001	0.342 ± 0.005	0.375 ± 0.006	0.225 ± 0.002	0.424 ± 0.005